DSCI 6004 – Natural Language Processing Final Report

University of New Haven, MS Data Science, Spring 2026

Project Title: An RAG Symptom for Psychiatric Question Answering

Author: Alima Aqsai & Ellen Martin

ABSTRACT

Psychiatric disorders are some of the hardest conditions to diagnose accurately and clinicians often have to work through vast amounts of clinical literature while managing complex, overlapping patient presentations. This paper describes a Retrieval-Augmented Generation (RAG) system built to support psychiatric diagnosis and treatment planning by grounding language model responses in real clinical documents, including the DSM-5, ICD-11, NIMH guidelines, and WHO publications. We compared three open-source models, Mistral-7B, LLaMA-3.1-8B, and Gemma 2, all running through the same pipeline, tested against 17 questions covering diagnosis, symptom differentiation, treatment, and comorbidity. Performance was measured using BLEU and ROUGE-L for lexical accuracy, roBERTScore for semantic similarity, Precision@k, Recall@k, and MRR for retrieval quality, and human ratings for clinical relevance and accuracy. All three models scored much higher on ROUGE-L than BLEU (ROUGE: 0.21–0.25 vs. BLEU: 0.03–0.09), which suggests they captured the right clinical concepts but expressed them in their own words rather than quoting the source directly. Mistral-7B performed best overall, with a mean BERTScore F1 of 0.860, compared to 0.848 for LLaMA and 0.845 for Gemma. Retrieving three chunks (k=3) was enough to capture the most relevant context, with F1@k peaking around k=7 and MRR settling near 0.6. Overall, the results show that open-source RAG systems can work reasonably well for psychiatric support, though expanding the document base and refining evaluation methods would strengthen future versions.

1. INTRODUCTION

Psychiatric disorders are a major and growing global health problem. Conditions like Major Depressive Disorder, Bipolar Disorder, Generalized Anxiety Disorder, and personality disorders affect hundreds of millions of people, yet getting a timely and accurate diagnosis remains genuinely difficult. Clinicians have to work through overlapping symptoms, frequent comorbidities, and a constantly evolving evidence base — often with limited time and resources. In many parts of the world, psychiatric specialists are simply not available, which makes access to good care even harder.

Large language models have attracted a lot of interest as tools that could help address some of these gaps. They can process and summarize large amounts of clinical text quickly, which makes them appealing for tasks like answering diagnostic questions or synthesizing treatment guidelines. The problem is that general-purpose LLMs hallucinate, they produce responses that sound plausible but are factually wrong, which is a real risk in a clinical setting. Retrieval-Augmented Generation (RAG) helps with this by pulling in relevant passages from trusted source documents before generating a response, so the model is working from real content rather than relying entirely on what it picked up during training.

Others have already started exploring this space. Wan et al. (2024) found that LLM-assisted nursing support led to better patient satisfaction and higher-quality responses. Wang et al. (2025) made the case that RAG is especially well-suited to medicine because it lets experts trace responses back to their sources. Closer to our domain, Cremaschi et al. (2025) built an RAG system called LLMind-Chat on ICD-11 content for psychiatric decision support, achieving an F1-score of 0.524 against a human clinician baseline of 76.1%. What is still missing is a direct comparison of multiple open-source models within the same psychiatric RAG setup.

That is what this project does. We built a single RAG pipeline and ran Mistral-7B, LLaMA-3.1-8B, and Gemma 2 through it, evaluating each on 17 questions designed to reflect real clinical reasoning in psychiatry. Source documents included the DSM-5, ICD-11, NIMH and NHS guidelines, open-access textbooks, and publications from the APA, BPS, and WHO. Chunks were embedded using pubmedBERT and stored in ChromaDB for retrieval.

We assessed each model across lexical, semantic, retrieval, and human evaluation dimensions, using BLEU, ROUGE-L, roBERTScore, Precision@k, Recall@k, MRR, and author ratings for clinical accuracy and relevance. The goal was to get a full picture of where each model performed well and where it fell short, with the broader aim of understanding how useful open-source RAG systems can realistically be for psychiatric clinical support.

2. METHOD

2.1 System Overview

The Retrieval-Augmented Generation (RAG) system that we built works in two steps: it first finds the most relevant passages from a library of clinical texts, then passes those passages to a language model to generate a response. Using retrieved content rather than relying purely on what the model learned during training helps reduce hallucination, which is a serious concern in clinical contexts.

We compared three open-source models, Mistral-7B-Instruct, LLaMA-3.1-8B-Instruct, and Gemma 2-9B-IT, all running through the exact same pipeline so that differences in performance reflect the model, not the setup. The system was built in Python using LangChain and run on Google Colab, with documents and outputs saved to Google Drive.

2.2 Data Sources

The document library consisted of five clinical sources chosen to cover the main areas of psychiatric diagnosis and treatment:

- DSM 5 TR.pdf: The DSM-5 Text Revision, the standard diagnostic manual for psychiatric disorders in North America.
- apa_treatment_depression.pdf: The APA's Clinical Practice Guideline for treating depression, covering both psychological and pharmacological approaches.
- fundamentals_of_psych_disorders.pdf: An open-access textbook on psychiatric classification, covering diagnostic limitations, comorbidity, and clinical decision-making.

- personality_disorders_dsm.pdf: A DSM-aligned reference focused specifically on personality disorder criteria.
- who_mental_health_gap.pdf: The WHO mhGAP Intervention Guide, focused on treatment in low-resource and non-specialized settings.

Together, these sources cover diagnostic frameworks, treatment guidelines, academic foundations, and global health perspectives, giving the system a broad and authoritative base to draw from.

2.3 Document Preprocessing and Chunking

Each PDF was loaded using PyMuPDFLoader, which extracts text page by page and keeps track of the source filename. The text was then split into chunks of 1,000 characters with a 100-character overlap between chunks. The overlap helps avoid losing information that falls at the boundary between two chunks.

One limitation to note: pubmedBERT, the embedding model we used, has a maximum input of 512 tokens. Some 1,000-character chunks may exceed this, meaning the end of longer chunks could be cut off silently during embedding.

2.4 Embedding and Vector Store

Chunks were converted into vector embeddings using **pubmedBERT** (microsoft/BiomedNLP-PubMedBERT-base-uncased-abstract-fulltext), which was pre-trained on biomedical literature and handles clinical language better than general-purpose encoders. All embeddings were stored in ChromaDB, which allows fast similarity-based search at query time. The vector store was built once and saved to Google Drive so it could be reused across all three model runs.

2.5 Large Language Models

We used three open-source instruction-tuned models, all loaded and run locally via HuggingFace Transformers. The table below shows their key specifications.

Table 1.

Model	Parameters	Developer	Architecture	Context Window	HuggingFace Identifier
Mistral-7B-Instruct-v0.1	7B	Mistral AI	Decoder-only Transformer	8,192 tokens	mistralai/Mistral-7B-Instruct-v0.1
LLaMA-3.1-8B-Instruct	8B	Meta AI	Decoder-only Transformer	128,000 tokens	meta-llama/Meta-Llama-3.1-8B-Instruct
Gemma 2-9B-IT	9B	Google DeepMind	Decoder-only Transformer	8,192 tokens	google/gemma-2-9b-it

All three models were loaded in 4-bit quantized format using BitsAndBytes to fit within Google Colab's memory limits. Generation settings were kept identical across models: maximum 512 new tokens, temperature 0.1 (for consistent, factual outputs), and sampling enabled with the end-of-sequence token as the pad token.

2.6 RAG Pipeline Configuration

Each query was embedded using pubmedBERT and matched against the ChromaDB store to retrieve the top 3 most similar chunks (k = 3). This retrieval depth was chosen based on preliminary testing and is supported by the Precision@k and MRR results in Section 5, which show performance levelling off at k = 3.

The retrieved chunks were concatenated and passed to the model alongside the question using LangChain's RetrievalQA chain in 'stuff' mode. The same prompt template was used for all three models, asking each one to answer using the provided context. This prompt is available in the Appendix (A1). Keeping the prompt the same across models meant any performance differences came from the model, not the prompt.

After generation, regular expressions were used to separate the answer from the sources section. The concatenated retrieved chunks were saved as the reference text for metric computation.

2.7 Evaluation Questions

We wrote 17 questions covering the main types of clinical reasoning in psychiatry: diagnosis, symptom differentiation, treatment planning, comorbidity, and the role of AI in care. The questions ranged from straightforward factual queries (e.g., diagnostic criteria for MDD) to applied vignettes requiring more integrative reasoning (e.g., determining a primary diagnosis when symptoms overlap).

Each question was assigned a gold source, the document most likely to contain the correct answer, which was used to compute retrieval accuracy metrics. The table below lists all 17 questions and their gold source assignments.

Table 2.

ID	Question	Gold Source
Q1	What are the core diagnostic criteria for Major Depressive Disorder, and how many symptoms are required for a formal diagnosis?	DSM-5 TR
Q2	How does Persistent Depressive Disorder differ from Major Depressive Disorder in terms of symptom duration and severity?	DSM-5 TR
Q3	What key features distinguish Generalized Anxiety Disorder from Panic Disorder?	DSM-5 TR
Q4	What are the essential impairments required for diagnosing a personality disorder?	personality_disorders_dsm
Q5	A patient reports persistent sadness, fatigue, and sleep disturbances lasting over two years. What is the most likely diagnosis, and what criteria support this conclusion?	DSM-5 TR
Q6	How can clinicians differentiate between depressive episodes in bipolar disorder and those in unipolar depression?	DSM-5 TR

Q7	A patient demonstrates impulsivity, lack of empathy, and manipulative behavior. What diagnosis should be considered, and what characteristics support it?	personality_disorders_dsm
Q8	What are the most effective psychological treatments for depression across different age groups?	apa_treatment_depression
Q9	How do psychotherapy and pharmacological treatments compare in effectiveness for treating depressive disorders?	apa_treatment_depression
Q10	What treatment strategies are recommended for managing depression in low-resource or non-specialized healthcare settings?	who_mental_health_gap
Q11	How should treatment plans be individualized based on patient-specific factors such as culture, comorbidities, and personal preferences?	apa_treatment_depression
Q12	How do diagnostic frameworks and treatment guidelines work together in clinical decision-making for psychiatric disorders?	fundamentals_of_psych_disorders
Q13	What are the limitations of symptom-based diagnostic approaches in psychiatry?	fundamentals_of_psych_disorders
Q14	How does comorbidity affect the accuracy of diagnosis and the effectiveness of treatment planning?	fundamentals_of_psych_disorders
Q15	Given a patient presenting with overlapping symptoms of anxiety and depression, how should a clinician determine the primary diagnosis and appropriate treatment approach?	DSM-5 TR
Q16	What challenges arise in accurately diagnosing psychiatric disorders, and how can clinicians address these challenges?	fundamentals_of_psych_disorders
Q17	How can AI-based systems assist in improving access to psychiatric diagnosis and treatment recommendations?	fundamentals_of_psych_disorders

2.8 Evaluation Framework

We evaluated each model across four dimensions: lexical similarity, semantic similarity, retrieval quality, and human judgment. The below table summarises all metrics used.

Table 3.

Metric	Category	Description
BLEU	Lexical	Measures n-gram precision between generated answer and retrieved reference text. Captures exact terminology overlap.
ROUGE-L	Lexical	Measures longest common subsequence recall between generated answer and reference. Captures clinical information coverage.
roBERTScore (Precision, Recall, F1)	Semantic	Computes token-level cosine similarity between contextual embeddings of generated and reference text using RoBERTa.
Precision@k	Retrieval	Proportion of retrieved chunks among the top-k that match the gold source document for each question.
Recall@k	Retrieval	Proportion of relevant gold source documents successfully retrieved within the top-k chunks.
MRR (Mean Reciprocal Rank)	Retrieval	Average reciprocal rank of the first correctly retrieved gold source document across all queries.

2.8.1 Lexical Metrics. BLEU and ROUGE-L measure how closely the model's wording matches the retrieved reference text. BLEU captures word-level precision (how much of the model's output appears in the reference), while ROUGE-L captures recall via the longest common subsequence (how much of the reference content made it into the response). The reference text was the concatenated retrieved chunks, not a manually written answer, so these scores reflect alignment with retrieved context rather than a perfect gold standard.

2.8.2 Semantic Metric. roBERTScore measures meaning-level similarity using contextual embeddings from a RoBERTa model, rather than counting shared words. This is useful because clinical language models tend to paraphrase rather than quote directly — roBERTScore captures whether the meaning was preserved even when the exact wording differs. Precision, Recall, and F1 were computed per question per model.

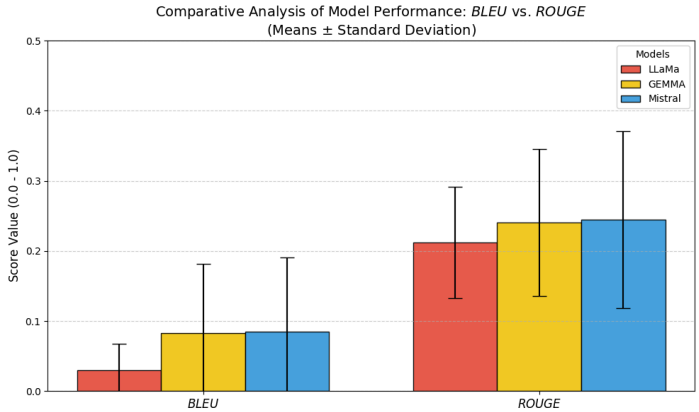
2.8.3 Retrieval Metrics. Precision@k and Recall@k were computed at k = 1, 3, 5, 7, 10, and 12 to evaluate whether the retrieval system was returning chunks from the correct source document. A chunk was counted as correct if its source matched the pre-assigned gold source for that question. MRR (Mean Reciprocal Rank) was also computed at each k to measure how highly the first correct document was ranked in the retrieved set.

3. RESULTS

3.1 Lexical Similarity: BLEU/ROUGE Scores

BLEU and ROUGE scores were computed for all three models (Mistral, LLaMa, Gemma) for each question, and on average, overall. The retrieved source chunks were used as the ground truth reference text for BLEU/ROUGE computations against the generated answer. BLEU captures exact-match precision, i.e., how closely the actual text generated matches the source material. In contrast, ROUGE captures how closely models capture the underlying information.

Figure 1. Comparative Analysis of Model Performance (BLEU vs ROUGE)



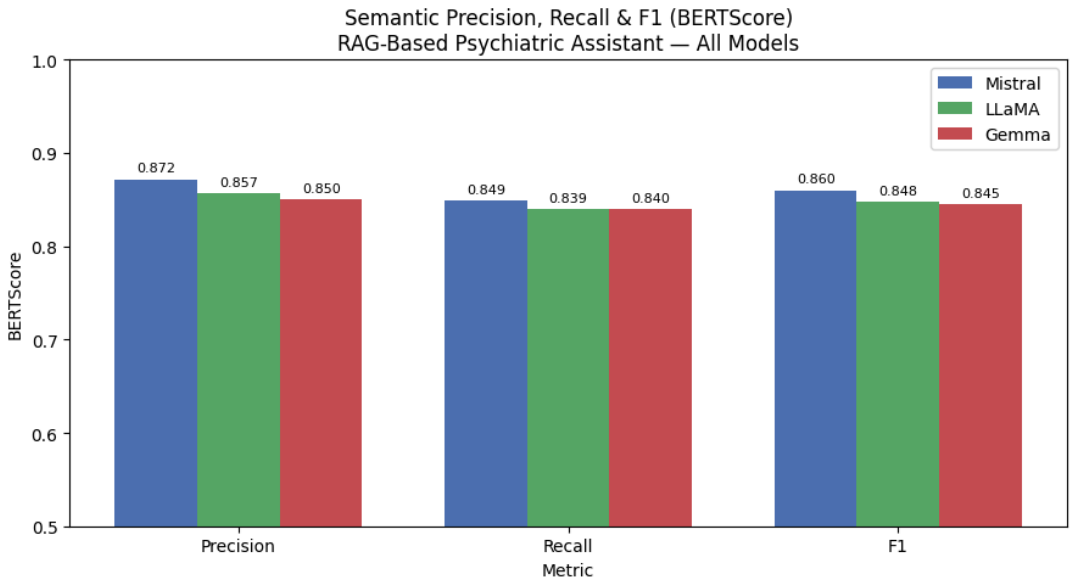
A one-way ANOVA found no statistically significant difference between BLEU scores across the three models, indicating that the models were approximately equivalent for exact-match precision. This was the same for ROUGE, suggesting that all three architectures perform equivalently at capturing the underlying clinical concepts in the source material. However, ROUGE scores were consistently higher than BLEU scores for each model, verified using paired t-tests ($p < 0.0001$ for all three models). LLaMA exhibited the largest difference

between BLEU and ROUGE, indicating that this model tends to reword information more than the other two architectures, while still maintaining the underlying clinical concepts.

3.2 Semantic Similarity: BERTScore

BERTScore was computed using the RoBERTa backbone to compute token-level semantic Precision, Recall and F1 performance. This was computed between the answers generated by the models and retrieved 'gold-standard' reference chunks. In this regard, beyond just comparing lexical similarity (like BLEU and ROUGE), BERTScore computes semantic similarity, checking that the RAG is capturing the correct meaning of the clinical/psychiatric text, rather than just using clinical/psychiatric terminology.

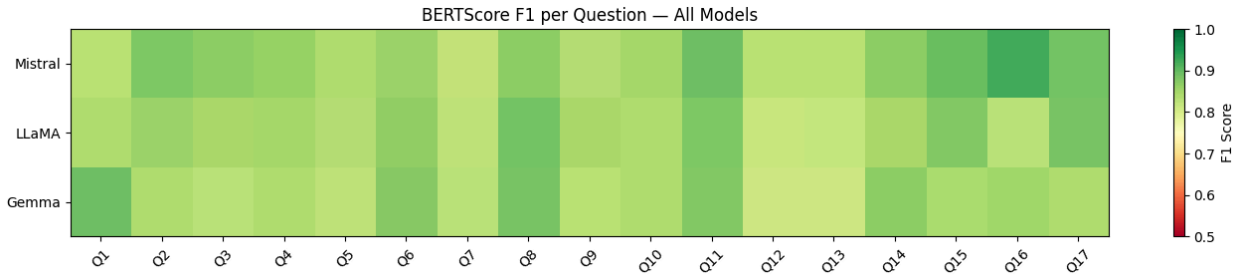
Figure 2. Semantic Precision, Recall & F1 (BERTScore)



Answers were truncated to 120 words prior to scoring to manage GPU constraints, which should be considered when evaluating the performance of the models, which sometimes produced long answers. All three models performed well according to F1. Mistral marginally outperformed both LLaMA and Gemma, but overall performance was similar across the three architectures. This suggests that the three models are capturing similar semantic information, in spite of some differences in how that information is conveyed in the model-generated answers. In

other words, differences between the models are likely a result of differences in phrasing and response structure, rather than differences in semantic meaning.

Figure 3. BERTScore F1 per Question

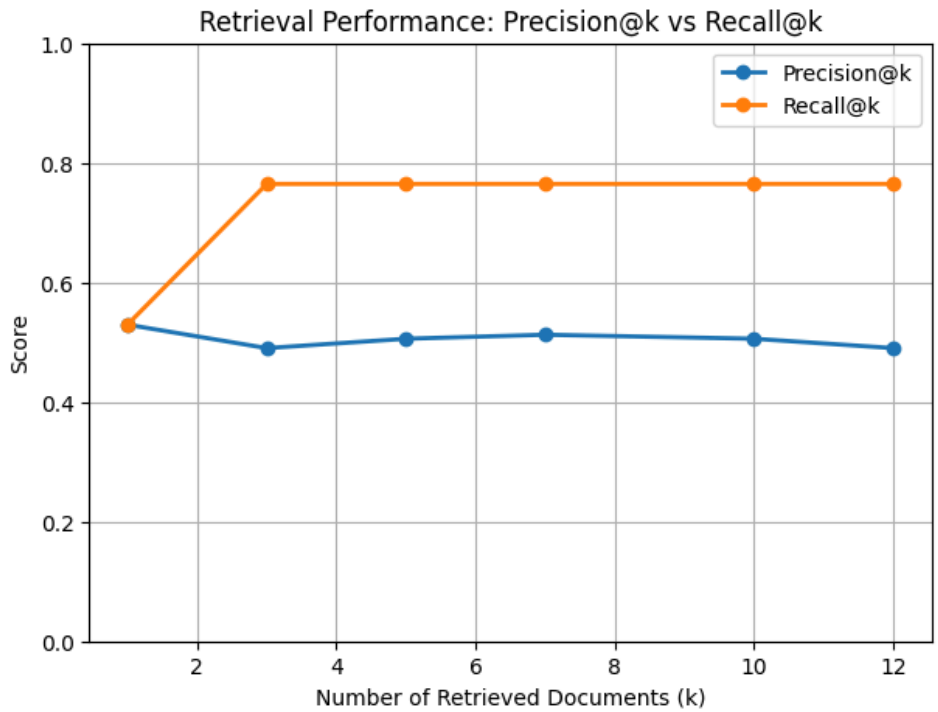


Comparing the F1 scores for each model for specific questions, Mistral seemed to achieve the best performance when answering question 16: What challenges arise in accurately diagnosing psychiatric disorders, and how can clinicians address these challenges? The answer Mistral provided is quite lengthy, and is shown in the Appendix (A2). Gemma seemed to achieve the best performance when answering question 1: What are the core diagnostic criteria for Major Depressive Disorder, and how many symptoms are required for a formal diagnosis? This was also a long response, and is shown in the Appendix (A3). Their elevated F1 scores may partially reflect how long the responses were for those particular questions which, because of their length, naturally contain more information and are more likely to achieve better semantic similarity.

3.3 Retrieval Evaluation: Precision@K, Recall@K

For each of the 17 questions, one of the source documents was used as the gold source document. Retrieval evaluation checked whether the chunks from the designated gold source appeared in the top-k retrieved results.

Figure 4. Precision@K vs Recall@K



Results found that recall performance plateaued at k=3 reaching a score of 0.764. This means that, for most questions, the gold-source document was retried within the top 3 chunks. Four questions had their correct source consistently missed regardless of how many chunks were retrieved, suggesting there is likely a gap in the knowledge base, rather than an issue with the models' architecture and performance as RAGs. This was intended, as we included one question on the applications of AI in psychiatry with the knowledge that this information was not explicitly included in any of the source material. Likewise, we included a question on complex comorbidity scenarios which the five source documents do not address in detail.

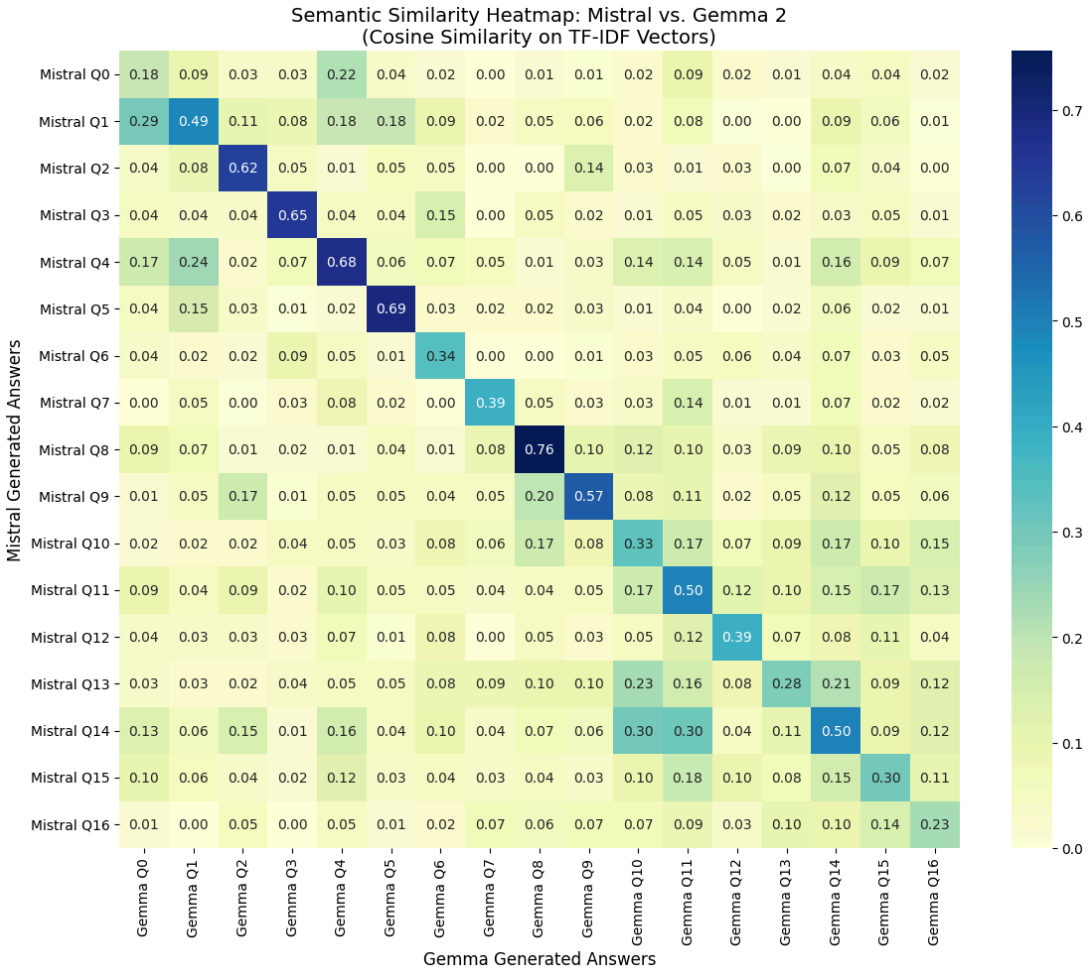
Precision peaked at 0.515 between k=8 and k=9. This highlights the trade-off between retrieving more chunks and retrieving high-quality chunks. Since the best source tends to already be reached by k=3, retrieving more k chunks may actually result in the retrieval of poorer-quality sources that are not relevant to the question.

3.4 Semantic Agreement between Mistral and Gemma

Since Mistral and Gemma marginally outperformed LLaMA, their answers were used to compute the semantic consistency of responses for the same questions. TF-IDF cosine similarity

was computed between Mistral- and Gemma-generated answers for each of the 17 evaluation questions, and then represented visually as a heatmap. High similarity scores on the diagonal suggest that, overall, the models are aligning in their responses to the 17 questions. Models showed the strongest agreement on treatment-focused questions, particularly when comparing psychotherapeutic vs pharmacological treatments and individualized care approaches.

Figure 5. Semantic Similarity Heatmap: Mistral vs Gemma (Cosine Similarity on TF-IDF)



There are, however, some discrepancies in the semantic consistency of responses, particularly to questions 1 and 17 (indices 0 and 16, respectively). For question 1 (symptoms required for depression diagnosis), Mistral's response is much more succinct, and does not provide as much detail as Gemma. Specifically, Gemma provides which symptoms are needed to fall into the mild, moderate and severe level of diagnosis, while Mistral provides a much more

general response (Appendix A4). Gemma also explicitly indicated that the source material did not indicate references to AI applications in psychiatry (for question 16), while Mistral still provided an answer, raising the question of whether its response is a hallucination or not.

4. DISCUSSION

Across almost every metric, all three models performed in a surprisingly similar range. None of them stood out as intensely better or worse than the others, which suggests that for this kind of task, answering psychiatric questions from a small, well-chosen document base, the architecture of the model matters less than how well the retrieval system is set up. Our findings closely resemble those in Eminovic et al. (2026), who compared Mistral, LLaMA and Gemma, among other models, in summarizing information from medical radiology reports on stroke patients, a somewhat similar task to question-answering using retrieval. They similarly found that those three models did not significantly differ in terms of BLEU/ROUGE or BERTScore, although their scores for BLEU/ROUGE were, on average, higher than those we computed.

We found that ROUGE was consistently higher than BLEU for all models, as did Eminovic et al. (2026). This pattern of results is generally a good sign for clinical natural language processing tasks, as it indicates that the underlying psychiatric information is being captured, but in a more conversational mode of language. An effective RAG system must be able to capture the correct psychiatric information, but it is not helpful for such a system to simply reproduce the exact text, particularly regarding dense diagnostic clinical manuals like the DSM-5.

4.1 Qualitative Distinctions Between Models

There were some qualitative differences between the model-generated answers. Gemma's responses were considerably longer than both Mistral's and LLaMa's and used bullet points in several answers. One key distinction noted in Gemma's responses for certain questions outside the scope of the reference material (i.e., the question on AI applications in psychiatry) was that Gemma explicitly stated that the source material did not contain information pertaining to that topic. Mistral and LLaMA both, notably, provided some attempt to answer that question, raising concerns of potentially greater hallucination in those models compared to Gemma. While recent work by Nadeu et al. (2024) does support our suggestion that LLaMA may have a high propensity to hallucinate, this work found that Mistral hallucinated the least, which somewhat contradicts our findings, although it should be noted that we did not formally test hallucination.

One reason for Gemma's more conservative and transparent responses could be the fact that DeepMind trains models in a way that encourages expressing levels of confidence in responses and simply declining to provide an answer if there is insufficient information available (Riviere et al., 2024). Mistral, contrastingly, prioritizes instruction-following and completing responses, even under uncertainty. In other words, Gemma and Mistral may have different reward policies as part of the Reinforcement Learning from Human Feedback (RLHF) process, encouraging transparency and completeness to different respective degree. Differences in sliding window length and attention mechanisms may also contribute to a greater understanding, in Gemma, of the immediate context information, as well as an understanding of what is outside of the scope of the current context, compared to Mistral (Jiang et al., 2023).

Similarly, one reason for LLaMA's lower BLEU score, which implies greater re-wording of source material, is likely attributable to the fact that LLaMA's instruction tuning phase involved heavy emphasis on RLHF as well as Direct Preference Optimization (DPO) (Dubey et al., 2024). As a result, it may have adopted a more generally conversational style, one that received more positive feedback during RLHF, as opposed to dense, highly technical phrasing like that found in the DSM-5 manual. LLaMA also uses a much larger vocabulary than Mistral (128,000 vs 32,000 tokens), meaning that it may just have a larger vocabulary to draw from when generating output, leading to greater deviation from the source material in terms of exact token matching. Furthermore, LLaMA may be better at representing information more holistically because of the Rotary Positional Embedding (RoPE) scaling, which captures long-term token dependencies better.

4.2 Methodological Considerations

One strength of this project is its use of multiple measures of performance evaluation. BLEU/ROUGE are good indicators of lexical/stylistic similarity, confirming that the RAG is not simply regurgitating complex material from diagnostic manuals, which is not necessarily helpful for the task of question-answering. Likewise, BERTScore is a good indicator of performance for psychiatric tasks, particularly. This is because psychiatric disorders and symptoms can be described in a multitude of ways (e.g., 'depressed' vs 'feeling sad' vs 'low mood' vs

‘anhedonia’), so an evaluation method that measures semantic matching, beyond exact matching, is likely to be more informative of performance.

One notable limitation of our methodology is our lack of external verification by a clinician. As explained by Muhetaer et al. (2025), “lexical and semantic alignment does not necessarily imply sufficient specificity, safety, or actionability”. Thus, while BLUE/ROUGE and BERTScore provide some measure of lexical and semantic similarity, there is limited insight into how clinically relevant and appropriate the model-generated responses were. Including clinician ratings of relevance and appropriateness would likely add great value to this project, but was out of the present scope.

CONCLUSION

This project set out to answer a practical question: can open-source language models, connected to a small library of clinical documents via a RAG pipeline, give useful and accurate answers to psychiatric questions? The short answer is yes, to a reasonable degree — but with some important caveats.

All three models — Mistral-7B, LLaMA-3.1-8B, and Gemma 2 — successfully recalled relevant clinical content from the retrieved chunks, producing responses that were semantically meaningful and largely free from clinical errors. Mistral performed best overall across automated metrics, but the differences between models were small enough that the choice of model matters less than the quality of the retrieval setup. Getting the right chunks into the model's context window was more important than which model processed them.

The clearest limitations were in the document base and the evaluation setup. Five source documents is a reasonable starting point, but it leaves obvious gaps — questions about AI in psychiatry, complex comorbidity scenarios, and topics not well covered by the DSM or APA guidelines consistently returned poor retrieval results. Expanding the corpus with more diverse and up-to-date sources would likely improve performance more than switching models. On the evaluation side, having the authors rate their own outputs without blinding or inter-rater reliability checks is a limitation that should be addressed in future work.

The chunk size mismatch between our 1,000-character chunks and pubmedBERT's 512-token input limit is also worth fixing. Silent truncation at the embedding stage means some chunks were not fully represented in the vector store, which may have hurt retrieval performance for longer, denser passages.

Looking ahead, there are several directions that would meaningfully improve a system like this. A larger and more varied document corpus — including ICD-11, NICE guidelines, and recent psychiatric research — would reduce knowledge gaps. Longer-context embedding models would handle larger chunks without truncation. Adding a re-ranking step after initial retrieval could improve the quality of what gets passed to the model. And independent clinical evaluation,

ideally by practicing psychiatrists, would give a much more reliable picture of real-world usefulness than author ratings can.

Despite these limitations, the results are encouraging. Open-source RAG systems are accessible, transparent, and customisable in ways that proprietary systems are not — and this project shows they can produce clinically grounded responses to psychiatric questions without requiring fine-tuning or expensive infrastructure. With a more robust document base and a stronger evaluation framework, a system like this could realistically serve as a useful support tool for clinicians, students, and researchers working in psychiatric care.

REFERENCES

Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., ... & Ganapathy, R. (2024). The Llama 3 herd of models. arXiv preprint arXiv:2407.21783.

Eminovic, S., Lindholz, M., Dell'Orco, A., Bendszus, M., & Pfaff, J. (2026). Comparing performance of seven fine-tuned open-source large language models in summarizing and predicting outcome-relevant information from mechanical thrombectomy reports in patients with acute ischemic stroke. *European Radiology*, *36*, 3661–3675. https://doi.org/10.1007/s00330-025-12122-x

Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., de las Casas, D., ... & El Sayed, W. (2023). *Mistral 7B*. arXiv preprint arXiv:2310.06825.

Muhetaer, M., Yusupu, A., Yifan, W., Mutalipu, M., & Hao, F. (2025). Medical QA dialogue datasets in RAG systems performance evaluation and ChatGPT optimization. *Scientific Reports*, *15*, Article 44467. https://doi.org/10.1038/s41598-025-28015-4

Nadeau, D., Kroutikov, M., McNeil, K., & Baribeau, S. (2024). *Benchmarking Llama2, Mistral, Gemma and GPT for factuality, toxicity, bias and propensity for hallucinations*. arXiv. https://doi.org/10.48550/arXiv.2404.09785

APPENDIX

A1. Prompt Template

```
Use the following pieces of context to answer the question at the end.

For each piece of context, the source filename is provided.

When you answer, try to mention which document the information came from.

At the very end of your answer, provide a section labeled 'SOURCES USED:'
and list the filenames.
```

```
Context:

{context}


Question: {question}


Helpful Answer:
```

A2. Mistral-generated response to question 16:

Accurately diagnosing psychiatric disorders can be challenging due to several factors. First, symptoms that are indicative of a diagnosis should not be used to justify limiting access to appropriate care (APA, 2022). This means that not all symptoms may be present in every case, and a diagnosis should not be withheld based on this.

Second, symptoms that cluster together regularly are called a syndrome, and if they also follow a predictable course, they are characteristic of a specific disorder. However, the classification systems used by mental health professionals to diagnose disorders are based on distinct categories, and people may present with symptoms that do not fit neatly into these categories (APA, 2022).

Third, cultural differences can affect the presentation of symptoms. For example, nonverbal communication, such as eye contact and body language, can vary across cultures. If the assessment is conducted in a language that is different from the individual's primary language, alogia (poverty of speech) may be related to linguistic barriers rather than a symptom of a disorder (APA, 2022).

Fourth, misdiagnosis of schizophrenia is more likely to occur in members of underserved ethnic and racialized groups, potentially due to clinical bias, racism, or discrimination leading to limited quality of information and potential misinterpretation of symptoms (APA, 2022).

Lastly, sex- and gender-related diagnostic issues can also arise. For example, the APA's Guideline for the Treatment of Depression asserts that clinicians seeking to optimize the patient's therapeutic journey are faced with a distinct challenge to meet the patient "where s/he stands." This means that clinicians must personalize treatment approaches, prioritize shared decision-making, communicate openly and directly (when appropriate), utilize technologies that improve accessibility (when viable), make a concerted effort to accept more insurance options or provide sliding-scale fee structures, and ultimately by creating a nurturing and strong therapeutic alliance and treatment engagement (APA, 2022).

To address these challenges, clinicians can use empirically guided judgment to personalize treatment approaches, prioritize shared decision-making, communicate openly and directly (when appropriate), utilize technologies that improve accessibility (when viable), make a concerted effort to accept more insurance options or

A3. Gemma-generated response to question 1

Episode are present but full criteria are not met, or there is a period lasting less than 2 months without any significant symptoms of a major depressive episode following the end of such an episode.

In full remission: During the past 2 months, no significant signs or symptoms of the disturbance were present.

Specify current severity: Severity is based on the number of criterion symptoms, the severity of those symptoms, and the degree of functional disability.

Mild: Few, if any, symptoms in excess of those required to make the diagnosis are present, the intensity of the symptoms is distressing but manageable, and the symptoms result in minor impairment in social or occupational functioning.

Moderate: The number of symptoms, intensity of symptoms, and/or functional impairment are between those specified for ,Äúmild,Äù and ,Äúsevere.,Äù

Severe: The number of symptoms is substantially in excess of that required to make the diagnosis, the intensity of the symptoms is seriously distressing and falling into distinct categories for which there are clear descriptions and criteria for making a diagnosis.

Elements of a diagnosis in the DSM include the diagnostic criteria and descriptors, subtypes and specifiers, the principle diagnosis, and a provisional diagnosis.

Section 3.2 Review Questions

What is clinical diagnosis?

1.What is a classification system and what are the two main ones used today?

2.Outline the diagnostic categories used in the DSM-5-TR.

3. Additional Resources and Future Directions

Conditions for Further Study

Described in Section III, ,ÄúConditions for Further Study,,Äù are those proposed conditions for which sufficient scientific evidence is not yet available to support widespread clinical use. These

proposed criteria and supporting text are included to highlight conditions that could benefit from further research.

Assessment and Monitoring Tools: Now and Looking to the Future

The various components of DSM-5 are provided to facilitate clinical assessment and to aid in developing a comprehensive case formulation (see ‚ÄúUse of the Manual‚Äù) and identification of characteristics that can influence the prognosis of any diagnosed mental disorders. Whereas the diagnostic criteria in Section II are well established and have undergone extensive review, the

A4. Mistral-generated response to question 1

The core diagnostic criteria for Major Depressive Disorder, as outlined in the DSM-5-TR, include at least five of the following symptoms: depressed mood, loss of interest or pleasure, significant weight loss or gain, insomnia or hypersomnia, psychomotor agitation or retardation, fatigue or loss of energy, feelings of worthlessness or excessive or inappropriate guilt, diminished ability to think or concentrate, or recurrent thoughts of death or suicide. For a formal diagnosis, at least five of these symptoms must be present nearly every day for a period of at least two weeks.